Biological Information Flow Ontology and Propagation Framework

Annotation Specification

BIFO v0.02

Biological systems are modeled as a multi-layered, stateful network in which entities carry internal state and interact through biologically admissible, directional transformations. Each transformation encodes a mechanistically grounded transition that respects biological causality, spatial constraints, energetic feasibility, and physical state. Information flow is defined as the propagation of state across entities under these constraints. Directionality of all transformations implies an intrinsic temporal arrow such that all state transitions follow *state(t) → state(t+1)*.

This framework enables representation of intracellular processes, intercellular communication, metabolism, energetics, electrochemical signaling, spatial organization, temporal progression, and phenotypic outcomes within a unified and mechanistically grounded system.

BIFO distinguishes between information-carrying flows, which propagate state, and constraint-imposing flows, which restrict admissible transitions.

Theoretical Position

BIFO addresses a gap not covered by existing frameworks. Ontologies such as GO and HPO provide semantic representations of biological entities and relationships. Standards such as SBML encode explicitly defined mechanistic models. Causal assertion frameworks such as GO-CAM and BEL describe relationships between biological activities. None of these define rules for how information is *allowed to propagate* through integrated, heterogeneous graph structures.

BIFO is a propagation theory: it defines the channel conditions under which biologically valid information flow occurs across any knowledge graph that satisfies the BIFO interface contract. Analogously to how Shannon's information theory defines channel capacity independently of the physical transmission medium, BIFO defines admissible biological information flow independently of the underlying knowledge graph implementation. The asymmetry introduced by admissible directional flows over a property graph is precisely what constrains traversal to semantically valid biological paths, adding information by restricting the reachable state space.

Graph Substrate Assumption

BIFO operates over any **property graph** with a node-edge-node (triple) construction in which:

- Nodes represent biological entities carrying typed internal state.

- Edges represent relationships between entities. Edges may themselves represent biological processes (e.g., enzymatic reactions, signaling events, transcriptional programs) rather than merely structural links.
- Node and edge types carry sufficient semantic information to be assigned to BIFO entity classes and flow classes respectively, either directly or through CURIE resolution (see Section: CURIE Mapping Layer).
- The graph supports provenance metadata sufficient to evaluate evidence type and confidence.

BIFO does not assume any specific graph schema, identifier system, or database implementation. It does not require bidirectional edges, UMLS CUI structure, or SAB-tagged provenance. Any property graph satisfying the above requirements is BIFO-compatible. A graph satisfying them partially is BIFO-compatible for the subset of flow classes whose requirements are met.

When an edge node is itself a biological process, the admissibility function must evaluate the process node's state in addition to the endpoint node states. This applies particularly to reaction-level graphs derived from metabolic network reconstructions or pathway databases.

Formal System Representation

The system can be expressed formally as:

$$s_v^{t+1} = \Sigma_{u \in N(v)} \; A_r(s_u^t, s_v^t, m, \ell) \cdot T_{r+u=v-}(s_u^t, s_v^t, m, \ell)$$

where:

- T_r is the transformation operator for flow class r
- m represents optional metabolic or biochemical model context
- ℓ represents spatial context
- A_r is the admissibility function

The admissibility function A_r determines whether a transformation is permitted under the current biological, spatial, and resource constraints. In a BIFO-conditioned graph, A_r is evaluated against: (1) the BIFO entity class of the source node, (2) the BIFO entity class of the target node, (3) the BIFO flow class of the connecting edge, and (4) the current state of both nodes. All inputs are defined in terms of BIFO's own type system, populated by CURIE resolution or direct type assignment depending on the KG implementation.

When metabolic or energetic reasoning is invoked, *m* denotes the subset of biochemical network structure and states required to preserve consistency with the relevant metabolic model. In analyses that do not invoke metabolic modeling, this term may be omitted.

The system consists of an informational state space defined over G, CH, RNA, P, CM, PW, C, PH, and DS, and a resource and physical state space defined over SM, ION, and MECH. Informational flow drives system behavior, while resource and physical states constrain admissible transitions.

All graphs to use BIFO need the following: CURIEs on typed nodes and edges, and provenance of assertions.

A. Specifications for Entities and States

Table 1: Entity classes represent the major organizational layers of biology.

Class	Description
G	Genetic sequence and variation (genes, loci, variants)
CH	Chromatin state over genomic regions
RNA	Transcripts with role-dependent function
P	Proteins with state-dependent activity
CM	Macromolecular complexes and assemblies
SM	Small molecules with role-dependent semantics
ION	Ionic and electrochemical state
PW	Pathways and biological programs
C	Cellular configuration. Cellular entities may represent individual cells or aggregated cellular populations depending on modeling resolution.
S	Spatial context across scales
PH	Phenotype (observable traits)
DS	Disease (multi-phenotype construct)
X	External perturbations
MECH	Mechanical and physical state

Table 2: State Definitions. Each entity carries an internal state appropriate to its biological role.

Entity	State Definition
CH	(accessibility, methylation, histone_modifications, 3D_topology)
RNA	(role ∈ {coding, regulatory, processing, structural}, abundance, stability, translation_state, processing_state, location)
P	(identity, PTM_state, activity, location)
CM	(composition, activity_state, location)
SM	(identity, role ∈ {energetic, metabolic, signaling, perturbational, environmental}, concentration, redox_state)
ION	(species, concentration, gradient, membrane_potential)
C	(chromatin_attractor (cell type for example), functional_state, cell_cycle_state)
S	(macro_anatomy, subcellular_compartment, extracellular_region, boundary_role)

Entity	State Definition
MECH	(force, stiffness, tension, geometry)
X	(identity, mechanism ∈ {inhibitor, activator, degrader, modulator, environmental}, target_entity, concentration_range, evidence_source)

NOTE: Binding interactions in BIFO are implicitly defined over sequence- or structure-specific substrates (e.g., genomic regions, RNA sequences, or protein interfaces), which determine admissibility of regulatory and signaling flows, including constraints imposed by spatial localization. Spatial context S forms a hierarchical and connected topology across anatomical regions, subcellular compartments, and extracellular domains, enabling constraint of transport, processing, and signaling flows.

Complex boundary criterion: A macromolecular complex (CM) is instantiated when two or more protein entities associate with sufficient stability to execute a coordinated function as a unit, evidenced by co-purification, structural data, or functional assay. Transient interactions that do not meet this criterion are represented as direct state transformations between P entities without persistent CM instantiation. When directionality evidence is available (e.g., one protein acts as kinase and the other as substrate), a transient interaction is represented as a directed Signal Transuction flow P(state) → P(state') rather than a symmetric Protein Interaction.

B. CURIE Mapping Layer

BIFO provides a two-level CURIE mapping architecture that enables KG-agnostic operation. Any knowledge graph whose nodes and edges carry resolvable CURIEs from standard biomedical ontologies can be BIFO-conditioned through identifier resolution, without requiring schema-specific mappings. BIFO compatibility can be computed as the fraction of nodes and edges in a given KG that successfully resolve through these mapping tables.

B.1 Level 1: CURIE-to-Entity-Class Mapping

The following ontology namespaces and identifier types map to BIFO entity classes. This mapping is normative and KG-agnostic.

BIFO Entity Class	Ontology / Namespace	Example CURIEs / Notes
G	HGNC, Ensembl gene (ENSG), NCBI Gene, Sequence Ontology gene features	HGNC:11998, ENSG00000141510, SO:0000704
G (variant)	ClinVar, dbSNP, gnomAD, Sequence Ontology variant types	rs12345, SO:0001060 (sequence variant)
CH	Sequence Ontology chromatin features, ENCODE terms, 4DN ontology	SO:0001747, histone modification terms
RNA	Ensembl transcript (ENST), RNAcentral, Sequence Ontology transcript features	ENST00000269305, SO:0000233
P	UniProtKB, Ensembl protein (ENSP), PRO ontology	P04637, ENSP00000269305, PR:000000001

BIFO Entity Class	Ontology / Namespace	Example CURIEs / Notes
CM	GO cellular component complexes, Complex Portal, IntAct complex	GO:0005667, CPX-123
SM	ChEBI, PubChem, HMDB, KEGG Compound	CHEBI:15422, CID:5793, HMDB0000122
ION	ChEBI ionic species, GO ion transport terms	CHEBI:29101 (sodium ion)
PW	Reactome, GO biological process, KEGG Pathway, WikiPathways	R-HSA-212436, GO:0006096, hsa00010
C	Cell Ontology, Azimuth cell types, CL terms	CL:0000057, CL:0000236
S	UBERON, FMA, GO cellular component (compartment)	UBERON:0002107, GO:0005737 (cytoplasm)
PH	HPO, OMIM phenotype, MP (mammalian phenotype)	HP:0001636, MP:0010403
DS	MONDO, DOID, OMIM disease, Orphanet	MONDO:0004975, DOID:14566
X	PubChem (drugs/compounds), DrugBank, ChEBI perturbational, LINCS perturbagen IDs	PUBCHEM:2247, DB00945
MECH	GO mechanical stimulus terms, SPARC ontology terms	GO:0009612

B.2 Level 2: Predicate-to-Flow-Class Mapping

The following Relations Ontology (RO) terms and common biomedical predicates map to BIFO flow classes with the directionality indicated. Where the same predicate maps to different flow classes depending on entity type, the entity-type-dependent rule takes precedence. This mapping is normative.

Predicate / RO Term	BIFO Flow Class	Direction	Entity-Type Condition
RO:0002292 (expressed in)	Transcription	G + CH → RNA	Subject: G or CH; Object: S or RNA
RO:0002434 (interacts with)	Protein Interaction	P ↔ P	Both entities: P
RO:0002213 (positively regulates)	Signal Transduction or Pathway Contribution	P → P' or P → PW	Depends on object entity class
RO:0002212 (negatively regulates)	Signal Termination or Negative Feedback	P → P_0 or downstream → upstream	Depends on object entity class
RO:0002233 (has input)	Biochemical Transformation or Metabolic Coupling	SM → SM or P ↔ SM	Reaction context
RO:0002234 (has output)	Biochemical Transformation	SM → SM	Reaction context

Predicate / RO Term	BIFO Flow Class	Direction	Entity-Type Condition
RO:0002411 (causally upstream of)	Any directed flow	Source → Target	Resolved by entity class
RO:0002304 (causally upstream of, positive effect)	Signal Transduction or Transcription	Source → Target	Resolved by entity class
BFO:0000066 (occurs in)	Spatial constraint (admissibility)	—	Contributes to S context
RO:0001025 (located in)	Intracellular Transport or Spatial constraint	entity(loc_i) → entity(loc□)	Resolved by context
RO:0002430 (involved in)	Pathway Contribution	P or CM → PW	—
RO:0002092 (happens during)	Cell Cycle Constraint	cell_cycle_state → system state	—
regulates transcription (custom/GO)	Protein-DNA Regulation	P or CM + CH → RNA	Subject: P or CM
positively_regulated_by (LINCS)	Perturbational Effect	SM(perturbational) → RNA / P	Subject: SM or X
negatively_regulated_by (LINCS)	Perturbational Effect	SM(perturbational) → RNA / P	Subject: SM or X
molecularly_interacts_with	RNA-RNA Binding or Protein Interaction	Resolved by entity class	—
located_in (genomic, HSCLO-type)	Spatial constraint / Chromatin Topology	CH → S	Genomic context

NOTE: Predicate-to-flow-class assignment is context-dependent. When the same predicate connects entity types matching multiple flow classes, the more specific flow class (lower in the flow taxonomy hierarchy) takes precedence. Implementation guides for specific KGs should document any predicate mappings that deviate from or extend this normative table.

C. Flow Classes

Flow classes define admissible biological transformations over entities and states.

Example of using flow classes

The primary organizational backbone can follow a hierarchical flow from genetic sequence to phenotype to disease such as:

$$G + CH \rightarrow RNA \rightarrow P(state) \rightarrow PW \rightarrow C \rightarrow PH \rightarrow DS \quad (Eq\ 1)$$

Table: Central dogma transformations

Flow	Source → Target	Meaning
Transcription	G + CH → RNA(coding)	Chromatin-gated gene expression
Translation	RNA(coding) → P	Protein synthesis

Table: Chromatin regulation

Flow	Source → Target	Meaning
Chromatin Control	CH → RNA	Transcription gating
Epigenetic Modification	P or RNA → CH	Chromatin modification
Chromatin Topology	CH → CH	Three-dimensional genomic interactions

Table: RNA-mediated regulation

Flow	Source → Target	Meaning
RNA Regulation	RNA(regulatory) → RNA(coding)	Repression or degradation
RNA Processing	RNA(processing) → RNA(coding)	Splicing and maturation
Translation Support	RNA(structural) → P	Ribosomal and tRNA function
RNA-RNA Binding	RNA(regulatory) + RNA(target) → RNA(target')	Sequence-specific hybridization
RNA Silencing (degradation)	RNA(regulatory) → RNA(target')	Target degradation
RNA Silencing (repression)	RNA(regulatory) → RNA(target')	Translation inhibition
RNA Stabilization	RNA(regulatory) → RNA(target')	Increased stability
RNA Processing via RNA	RNA(processing) → RNA(target')	Editing or maturation

Table : Protein and complex signaling. Stable assemblies are represented as macromolecular complexes (CM), while transient interactions are represented as direct state transformations between proteins without persistent complex formation. See Section A for the formal complex boundary criterion.

Flow	Source → Target	Meaning
Signal Transduction	P(state) → P(state')	PTM-driven signaling
Transient Interaction Signaling	P(state) → P(state')	Short-lived signaling interaction
Complex Formation	P → CM	Assembly of complexes
Complex Function	CM → P or PW	Functional execution
Protein Interaction	P ↔ P	Binding interactions
Pathway Contribution	P or CM → PW	Participation in biological programs
Protein-DNA Regulation	P or CM + CH(region) → RNA(coding)	Transcription factor binding regulates transcription
Signal Termination	P(state) → P($state_0$)	Reversal or attenuation of signaling

Flow	Source → Target	Meaning
Negative Feedback	downstream → upstream	Regulatory feedback within signaling pathways

Table : Genetic modulation of transcription. Genetic regulatory modulation operates by altering sequence-dependent binding and chromatin accessibility, thereby modifying protein-DNA regulatory interactions.

Flow	Source → Target	Meaning
Genetic Regulatory Modulation	G(variant) → CH or RNA	Sequence variation alters transcriptional output

Table : Small-molecule transformations

Flow	Source → Target	Meaning
Biochemical Transformation	SM(metabolic) → SM(metabolic)	Substrate-product conversion
Metabolic Coupling	P or CM ↔ SM(metabolic)	Enzyme-mediated transformation
Energetic Constraint	SM(energetic) → system state	Contextual dependency on energetic cofactors and compatibility with metabolic models

Table : Redox and electron transfer

Flow	Source → Target	Meaning
Electron Transfer	SM(redox) → SM(redox)	Redox transitions
Electrochemical Coupling	SM(redox) → ION	Gradient generation
Chemiosmotic Coupling	ION → SM(energy)	ATP synthesis

Table : Small-molecule signaling and perturbation

Flow	Source → Target	Meaning
Molecular Signaling	SM(signaling) → P or RNA or CH	Endogenous signaling
Perturbational Effect	SM(perturbational) or X → system state	External modulation
Metabolic Regime Control	SM(environmental) → system state	Oxygen and carbon dioxide constraints
Nuclear Signaling	SM(signal) → P(state, location) → CH → RNA	Ligand-driven transcriptional activation
Receptor-Mediated Signaling	SM(signal) → P(receptor) → P(state')	Ligand binding activates receptor and initiates signaling cascade

Table: Electrochemical signaling

Flow	Source → Target	Meaning
Ion Channel Flow	P(state) → ION	Channel-mediated gradient change
Ionic Propagation	ION → ION	Spatial propagation of gradients
Electrochemical Signaling	ION → P or RNA or PW or C	Downstream signaling

Table : Spatial and transport processes

Flow	Source → Target	Meaning
Intracellular Transport	entity(loc_i) → entity(loc□)	Movement between compartments
Vesicular Transport	entity → entity	ER-Golgi-membrane trafficking
Nuclear Transport	entity ↔ nucleus	Import and export
Organelle Targeting	entity → organelle	Localization
Compartmental Processing	entity(loc) → entity(state', loc)	Compartment-restricted processing
Secretion	intracellular → extracellular	Export
Uptake	extracellular → intracellular	Import
Extracellular Transport	$region_i$ → region□	Diffusion or circulation
Source-Sink Exchange	source ↔ sink	Production and clearance
Molecular Degradation	entity → sink	Removal of molecules from system
Proteasomal Processing	P → degraded state	Ubiquitin-proteasome degradation
Autophagic Processing	CM / organelle → degraded state	Bulk degradation via autophagy

Table: Cell-cell communication

Flow	Source → Target	Meaning
Cell-Cell Communication	C_1 → C_2	Intercellular signaling

Table : Cellular state and progression

Flow	Source → Target	Meaning
Chromatin → Cell	CH → C	Defines attractor-like identity
Functional Transition	PW or P or SM → C	State transitions
Cell Cycle Control	P or PW → cell_cycle_state	Cell cycle regulation

Flow	Source → Target	Meaning
Cell Cycle Constraint	cell_cycle_state → system state	Phase-dependent gating
State Progression	state(t) → state(t+1)	Temporal evolution

Table : Mechanotransduction

Flow	Source → Target	Meaning
Mechanotransduction	MECH → P or CH or RNA or C	Physical signaling

Table : Phenotype and disease

Flow	Source → Target	Meaning
Cell → Phenotype	C → PH	Trait emergence
Molecular → Phenotype	P or PW → PH	Molecular effects
Phenotype Aggregation	PH → DS	Disease definition

Table : Observational and inference relationships. Observational relationships inform model construction and weighting but are not propagated as mechanistic flows unless explicitly validated. The mechanistic/observational distinction is load-bearing: flows in this table must not be used as propagation paths in BIFO-conditioned graphs. They may be used as priors, weights, or constraints on mechanistic flows, but do not themselves carry biological state. Examples of data types that produce observational rather than mechanistic edges include: single-cell clustering assignments, ATAC-seq accessibility correlations, spatial transcriptomics co-occurrence, multi-omic integration outputs, and topological graph proximity scores.

Flow Class	Source → Target	Meaning
Observational Inference	CH / RNA / S → C	Cell configuration inferred from multi-omic measurements
Observational Association	CH / RNA / P → PW / PH	Correlation-based mapping between molecular states and pathways or phenotypes
Spatial Co-localization	entity(loc_i) ↔ entity(loc□)	Co-occurrence in spatial measurements without causal implication
Multi-omic Integration	CH + RNA → PW / C	Joint inference from multiple modalities

D. Admissibility Constraints

Table : Admissibility constraints govern all transformations.

Constraint	Description
Directionality	All flows follow biological causality with implicit time dependence. Each directed flow is a distinct biological process; the forward and reverse of a given transformation are represented as separate flows with independent admissibility conditions, enzymes, and energetic requirements.
State dependence	Flows depend on elements such as CH, RNA role, P(state), SM, ION, MECH, and location.
Metabolic model consistency	For analyses invoking energetic or metabolic state, BIFO requires inclusion of the relevant genes, enzymes, metabolites, reactions, and compartmental states needed to preserve consistency with an underlying genome-scale metabolic model or related biochemical network representation.
Spatial constraints	Processes restricted to valid compartments. Spatial admissibility is evaluated against the S entity class; flows are inadmissible if source and target entities occupy incompatible spatial contexts.
Temporal progression	All transitions follow state(t) → state(t+1).
Non-flow exclusion	Remove hierarchy edges (is_a, part_of), identifier/lexical edges (CODE, TERM, label), and purely statistical or topological relationships. These must not be treated as information-carrying flows.
Contextual admissibility	Flow validity may depend on local context (e.g., compartment, cell type, spatial region) rather than global system state.

E. BIFO Graph Conditioning Protocol

The following four-step protocol converts a raw property graph into a BIFO-conditioned graph. This protocol is KG-agnostic and applies to any graph satisfying the requirements in the Graph Substrate Assumption section. KG-specific implementation guides document how each step is performed for a given graph.

Step 1: Entity Class Assignment

Assign each node a BIFO entity class from {G, CH, RNA, P, CM, SM, ION, PW, C, S, PH, DS, X, MECH} using the CURIE-to-entity-class mapping (Section B.1) or direct type assignment if the KG carries typed nodes. Nodes that cannot be assigned to any BIFO entity class are excluded from the conditioned graph. Record coverage as the fraction of nodes successfully assigned.

Step 2: Flow Class Assignment and Edge Filtering

For each edge, attempt assignment to a BIFO flow class using the predicate-to-flow-class mapping (Section B.2). Apply the following filters, removing edges that are:

- Hierarchy edges: is_a, part_of, subClassOf, and equivalent structural relationships
- Identifier and lexical edges: CODE, TERM, label, synonym, cross-reference, and related annotation links
- Purely statistical relationships: correlation edges, co-expression edges without directional evidence, topological proximity scores
- Edges connecting node pairs whose entity class combination does not match any admissible flow in the BIFO flow taxonomy

Edges that map to observational flow classes (Table 17) are retained in the conditioned graph but flagged as non-propagating. They may inform weighting or constraint evaluation but must not carry biological state in forward propagation.

Step 3: State and Spatial Annotation

Annotate nodes and edges with state and context information required for admissibility evaluation:

- Attach relevant state components to each node per Table 2 (CH state, RNA role, P(state), SM role, X mechanism, etc.) from available KG properties or ontology-derived defaults
- Assign spatial context (S entity class) to entities where location information is available, using the CURIE-to-entity-class mapping for GO cellular component, UBERON, or equivalent terms
- For flows involving the CM boundary criterion, apply the complex/transient classification rule defined in Section A

Step 4: Confidence and Provenance Annotation

Annotate each retained edge with confidence and provenance attributes per Table 19. Apply the following confidence propagation rule for composite multi-hop flows:

Confidence propagation rule: The confidence of a composite flow path is the minimum edge confidence along the path (weakest-link principle), weighted by evidence type. Evidence types are ordered: experimental > curated > inferred > predicted. When edges of mixed evidence type appear in a path, the composite evidence type is the lowest-ranked type present. Confidence scores from heterogeneous sources must not be compared directly; they should be normalized within source before composition.

F. Confidence and Provenance

Table 19: Confidence and provenance measures

Things like gene->RNA transcript as a “truth” condition might not need this but things like protein(state1)->protein(state2) might – do we bring in transcription factors or methylation etc etc.

Attribute	Description
Confidence	Strength or reliability of relationship, expressed as a normalized score within source or as a categorical level (high/medium/low)
Provenance	Source dataset or ontology from which the edge was derived
Evidence Type	Experimental vs inferred vs curated vs predicted
Composite Confidence	For multi-hop paths: minimum edge confidence along path, typed at the lowest evidence tier present

Implementation Guides

Implementation guides for specific knowledge graphs document how the BIFO Graph Conditioning Protocol (Section E) is applied to a given KG's schema, identifier system, and edge vocabulary. An implementation guide specifies: (1) the mapping from KG node types and identifiers to BIFO entity classes, (2) the mapping from KG edge types and predicates to BIFO flow classes, (3) which BIFO flow

classes are fully, partially, or not instantiable in the current KG build, and (4) any KG-specific extensions or restrictions to the normative CURIE mapping tables. Implementation guides are published and versioned separately from the BIFO specification. The DDKG-BIFO Implementation Guide serves as the reference implementation.